

SupportPilot — AI Ticket Management Agent

Dashboard Structure, Login / RBAC, Complete Ticket Flow & Testing Blueprint

This document defines the recommended frontend workspaces, login roles, ticket lifecycle, AI-agent flow, Milestone 1–4 connection, Jira/email integrations, escalation flow and practical test flows for the SupportPilot project.

Source basis: The project specification defines three core roles — customer, agent and admin — while the technical blueprint also provides internal workspaces for support staff, knowledge management, leads and AI operations. The architecture should therefore use one application with RBAC rather than many independent portals. ■filecite■turn4file1■ ■filecite■turn4file3■

1. How Many Dashboards / Workspaces Do We Need?

Recommended final structure:

1. CUSTOMER / EMPLOYEE PORTAL
2. SUPPORT AGENT CONSOLE
3. LEAD / ESCALATION WORKSPACE
4. KNOWLEDGE MANAGEMENT WORKSPACE
5. AI AGENT OPERATIONS CENTER
6. ADMIN CONSOLE
7. ANALYTICS & REPORTING

Important: these are logical workspaces inside ONE frontend application.
Do not create seven separate websites.

The technical blueprint explicitly defines a PortalLayout, ConsoleLayout and AdminLayout, and page groups for dashboard, tickets, AI agent, knowledge, escalations, analytics, integrations and administration. ■filecite■turn4file7■

2. Roles & Login

```
LOGIN
↓
Authentication
↓
JWT access/refresh token
↓
Load role + permissions
↓
Route guard
↓
CUSTOMER → /portal
AGENT → /tickets/queue
LEAD → /escalations / dashboard
KNOWLEDGE_MGR → /knowledge
ENGINEER → /ai-agent + tickets
ADMIN → /admin
MANAGER → /analytics
```

The simplified internship scope has three required roles: customer, agent and admin. The enterprise technical blueprint expands internal permissions to Agent, Engineer, Lead, Knowledge Manager and Admin. Implement the three required roles first, then add the internal roles through RBAC if the full blueprint is being built. ■filecite■turn4file10■ ■filecite■turn4file11■

3. Customer / Employee Portal

```
CUSTOMER LOGIN
↓
My Tickets Dashboard
■ Open Tickets
■ In Progress
■ Resolved
■ Closed
↓
Create New Ticket
■ Subject
■ Description
■ Category (optional)
■ Attachment
↓
Submit
↓
```

Ticket ID
↓
Track Status
↓
Receive email updates
↓
Confirm resolution / reopen when policy allows

A customer must only see their own tickets. This is an explicit security requirement in the project material.

■filecite■turn4file10■

4. Support Agent Console

AGENT LOGIN
↓
Agent Dashboard
■ ■ Priority Queue
■ ■ SLA Countdown
■ ■ My Assigned Tickets
■ ■ Unassigned Tickets
■ ■ High Priority
■ ■ Escalated
↓
Open Ticket
■ ■ Ticket details
■ ■ AI category / severity / priority
■ ■ Similar resolved ticket
■ ■ Suggested response
■ ■ Knowledge citations
■ ■ Activity history
■ ■ SLA status
↓
Agent Actions
■ ■ Claim / Assign
■ ■ Reply
■ ■ Edit AI suggestion
■ ■ Reclassify
■ ■ Escalate
■ ■ Resolve
■ ■ Close

The project specifically requires a smart priority-sorted queue with category/status filters and SLA countdown, followed by Open → In Progress → Resolved → Closed lifecycle tracking. ■filecite■turn4file1■

5. Lead / Escalation Workspace

LEAD LOGIN
↓
Escalation Board
↓
Team queues
■ ■ Billing
■ ■ Technical
■ ■ Account
■ ■ General
↓
At-risk / breached SLA
↓
Review escalation reason
↓
Reassign / prioritize / intervene
↓
Monitor resolution
↓
Close escalation

The technical blueprint includes an Escalation Board and SLA Monitor for Lead/Admin roles, including team-based escalation and breach monitoring. ■filecite■turn4file11■

6. Knowledge Management Workspace

KNOWLEDGE MANAGER / ADMIN
↓
Knowledge Dashboard
■ ■ Articles
■ ■ Create Article

■ Edit Article
■ Publish / Unpublish
■ Version History
■ Ingestion Status
↓
Document upload
↓
Chunking
↓
Embedding
↓
Vector index
↓
Available to M2 RAG
↓
Consumed by M3 Knowledge Retrieval Agent

M3 should reuse the M2 RAG capability rather than creating a second independent retrieval system. ■filecite■turn4file18■

7. AI Agent Operations Center

AUTHORIZED ENGINEER / ADMIN
↓
AI Agent Operations Center
■ Overview
■ Classification Agent / NLP
■ Diagnosis Agent
■ Knowledge Retrieval Agent
■ Resolution Agent
■ Validation / Confidence
■ Escalation Agent
■ Agent Runs
■ Failure / Latency Monitoring
↓
Agent Run Detail
■ Input
■ Retrieved chunks
■ Output
■ Confidence
■ Latency
■ Errors
■ Final decision

One AI Operations Center is enough. Do not create a separate portal for Diagnosis, Retrieval, Resolution and Escalation agents. The technical blueprint explicitly provides an Agent Workbench and Agent Run Detail pages. ■filecite■turn4file11■

8. Admin Console

ADMIN LOGIN
↓
Admin Console
■ Users
■ Roles & Permissions
■ Teams
■ SLA Policies
■ Routing Rules
■ AI Settings
■ Integrations
■ Jira
■ Email
■ Audit Logs
■ System Settings

These admin capabilities map directly to the technical blueprint's Users, Roles, Teams, SLA Policy, Routing Rules, AI Settings, Integrations and Audit Log pages. ■filecite■turn4file11■

9. Analytics & Reporting

MANAGER / LEAD / ADMIN
↓
Analytics Dashboard
■ Ticket Volume
■ Category Mix
■ Resolution Rate
■ AI Resolution / Deflection Rate
■ MTTR
■ SLA Breaches

- Agent Performance
- CSAT
- AI Quality
- Custom Reports

The blueprint defines Analytics Overview, Agent Performance, AI Quality and Report Builder pages. ■filecite■turn4file11■

10. Final Dashboard Count

MAIN WORKSPACES = 7

1. Customer / Employee Portal
2. Support Agent Console
3. Lead / Escalation Workspace
4. Knowledge Management Workspace
5. AI Agent Operations Center
6. Admin Console
7. Analytics & Reporting

For the simplified 40-day internship:

CORE = 3 roles

- Customer
- Agent
- Admin

Then add Lead / Knowledge / Engineer / Manager permissions as advanced internal workspaces if the full enterprise blueprint is implemented.

11. Complete Ticket End-to-End Flow

```

Customer / Employee
↓
Login
↓
Create Ticket
↓
Core API validates + saves ticket as NEW
↓
Async processing starts
↓
M1 Intelligence
■ Category
■ Sub-category
■ Severity
■ Priority
↓
M2 Knowledge Retrieval / RAG
■ Search KB
■ Retrieve relevant articles
■ Grounded context
↓
M3 Multi-Agent Orchestrator
■ Diagnosis Agent
■ Knowledge Retrieval Agent
■ Resolution Agent
■ Validation
↓
Confidence Gate
■ HIGH CONFIDENCE
■ ↓
■ AI Resolution
■ ↓
■ Email
■ ↓
■ Await user confirmation
■
■ LOW CONFIDENCE
↓
Escalation Agent
↓
Team Queue
↓
Jira
↓
Human Agent
↓
Agent resolves
↓

```

Open → In Progress → Resolved → Closed
↓
Activity Logs
↓
Analytics / SLA / CSAT

12. M1 → M2 → M3 → M4 Connection

M1 – Ticket Processing
Ticket intake
→ classification
→ severity
→ priority

↓

M2 – Knowledge + RAG
Current ticket
→ semantic/vector retrieval
→ knowledge context
→ grounded resolution

↓

- M3 – Multi-Agent + Integrations
 - M1 + M2 context
 - Orchestrator
 - Diagnosis
 - Retrieval
 - Resolution
 - Validation
 - Auto-resolution OR Escalation
 - Jira + Email + Activity Log

↓

- M4 – Dashboard + Optimization Analytics
 - monitoring
 - escalation/SLA
 - testing
 - optimization
 - deployment

This exact separation is important: M3 consumes M1/M2 outputs and does not duplicate M2. ■filecite■turn4file18■

13. M3 Multi-Agent Flow

[illegible]

The supplied M3 process specifically defines Diagnosis → Knowledge Retrieval → Resolution → Validation, followed by automated response or Escalation Agent. ■filecite■turn4file5■

14. Jira Integration Flow

Ticket ready for enterprise tracking
↓
Create Jira issue
↓
Save Jira issue ID/key
↓
SupportPilot ↔ Jira status synchronization

AI working
→ SupportPilot In Progress
→ Jira In Progress

Escalated
→ Assign support team
→ Jira assignee/team update

Resolved
→ Store resolution
→ Jira Resolved

Jira changes
→ Poll/webhook
→ Sync back to SupportPilot

The M3 guide specifies create/update/status synchronization and a Jira mapping record for each SupportPilot ticket. ■filecite■turn4file6■

15. Email Automation Flow

Ticket Created
→ Ticket Received email
→ Requester

AI Resolution Ready
→ Resolution email
→ Requester

Escalation
→ Escalation email
→ Requester + Support Team

Ticket Resolved
→ Resolved email
→ Requester

Every email
→ delivery status
→ email log
→ ticket activity history

These four notification types and email logging are specified in the M3 integration guide. ■filecite■turn4file6■

16. Ticket State Machine

NEW
↓
CLASSIFIED
↓
PRIORITIZED
↓
AI_PROCESSING
↓
AI_RESOLVED ■■■■■→ Await User Confirmation
■
■■ Low Confidence
↓
ESCALATED
↓
ASSIGNED
↓
IN_PROGRESS
↓

RESOLVED

↓

CLOSED

Optional:

REOPENED → IN_PROGRESS

17. Frontend Route Structure

```
app/
├── auth/
│   ├── login
│   ├── forgot-password
│   ├── reset-password
│   └── portal/
│       ├── my-tickets
│       ├── new-ticket
│       ├── ticket/:id
│       ├── self-help
│       ├── dashboard/
│       │   ├── overview
│       │   ├── tickets/
│       │   │   ├── list
│       │   │   ├── queue
│       │   │   └── :id
│       │   ├── ai-agent/
│       │   │   ├── workbench
│       │   │   ├── runs/:id
│       │   │   ├── knowledge/
│       │   │   │   ├── articles
│       │   │   │   ├── new
│       │   │   │   ├── :id/edit
│       │   │   ├── ingestion
│       │   │   ├── escalations/
│       │   │   │   ├── board
│       │   │   │   ├── sla
│       │   │   ├── analytics/
│       │   │   │   ├── overview
│       │   │   │   ├── agents
│       │   │   │   ├── ai-quality
│       │   │   │   ├── reports
│       │   │   ├── integrations/
│       │   │   │   ├── jira
│       │   │   │   ├── email
│       │   ├── admin/
│       │   │   ├── users
│       │   │   ├── roles
│       │   │   ├── teams
│       │   │   ├── sla
│       │   │   ├── routing
│       │   │   ├── ai-settings
│       │   │   └── audit
```

This route structure follows the technical frontend blueprint. [filecite turn4file7](#)

18. Login & RBAC Test Flow

Valid Customer

- login success
- Customer Portal
- can see own tickets only

Valid Agent

- login success
- Agent Queue
- can claim/update permitted tickets

Valid Admin

- login success
- Admin Console
- can manage configuration

Unauthorized Customer → /admin

- 403 / redirect

Unauthorized Customer → another customer's ticket

- 403 / 404

Unauthorized Agent → admin settings
→ 403

Expired JWT
→ refresh or login

Logout
→ protected pages inaccessible

19. Ticket Creation Test Flow

1. Login as customer.
2. Submit subject + description.
3. Attach optional screenshot.
4. Frontend validation passes.
5. Backend validation passes.
6. Ticket saved as NEW.
7. Ticket ID returned immediately.
8. Async processing starts.
9. Classification and priority are persisted.
10. Ticket appears in customer portal.
11. Ticket appears in agent queue.
12. Ticket-created email is logged/sent.

20. AI Classification Test

Input: "My account is locked and I cannot login."

Expected:
Category → Account
Priority → based on urgency rules/model
Confidence → stored

Failure test:
If classifier fails
→ safe fallback
→ ticket creation must still succeed
→ fallback category/priority recorded
→ error logged

The project explicitly states that AI failure must not prevent ticket creation; safe defaults should be used. ■filecite■turn4file14■

21. M2 RAG Test

Ticket
↓
Search knowledge base
↓
Retrieve relevant articles
↓
Verify source references
↓
Generate grounded context
↓
Resolution generation
↓
Store retrieved sources
↓
M3 receives the context

Negative:
No relevant article
→ low confidence
→ do not invent a solution
→ escalate or ask for human handling.

22. M3 Multi-Agent Test

Normal complex ticket
→ Orchestrator starts
→ Diagnosis runs
→ Retrieval runs
→ Resolution runs
→ Validation passes
→ automated response
→ email sent
→ Jira updated if required

→ activity log complete

Low-confidence ticket

→ validation fails

→ Escalation Agent

→ support team queue

→ Jira issue/update

→ escalation email

→ human agent resolves

23. Jira Test Flow

Create ticket ready for Jira

→ Create Jira issue

→ Verify Jira key saved

→ Change SupportPilot status

→ Verify Jira status

→ Change Jira status

→ Verify SupportPilot sync

→ Escalate

→ Verify Jira assignment

→ Resolve

→ Verify Jira Resolved

→ Verify last_updated timestamp.

24. Email Test Flow

Ticket Created → received email

AI Resolution Ready → resolution email

Escalation → escalation email

Resolved → resolved email

For each:

→ verify recipient

→ verify template

→ verify ticket ID

→ verify delivery/log status

→ verify activity log

25. SLA / Escalation Test

Create HIGH priority ticket

↓

SLA timer starts

↓

Agent queue shows countdown

↓

No response before threshold

↓

At-risk state

↓

Lead/Admin SLA Monitor

↓

Escalation rule

↓

Escalation Board

↓

Reassign / notify / Jira

↓

Audit event

26. Agent Failure Test

AI Service timeout

→ retry

→ if failure continues, return safe degraded result

→ do not lose ticket

→ do not fabricate resolution

→ log agent failure

→ show failed run in Agent Operations Center

→ escalate if confidence/availability is insufficient

27. Security Test Checklist

■ Passwords hashed

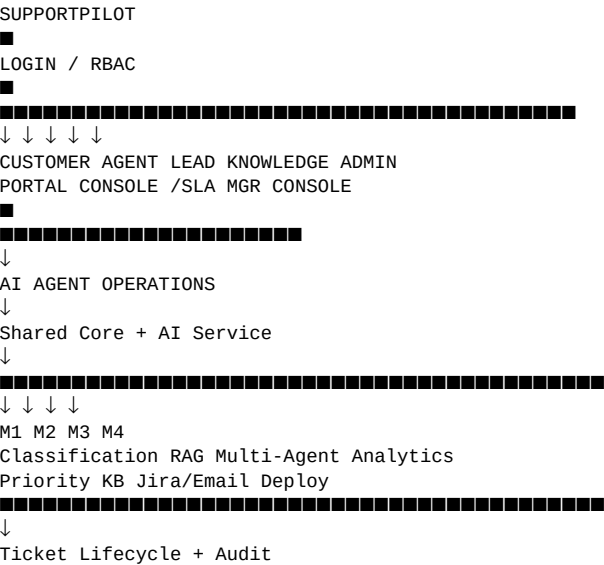
■ JWT expiry enforced

- Refresh token protected
- Role checks on every protected API
- Customer data isolated by owner
- Agent cannot access admin-only settings
- File upload validation
- Rate limiting / throttling
- CORS restricted
- Audit logs immutable
- Sensitive integration credentials protected
- Unauthorized ticket access rejected

28. Final Acceptance Checklist

- Customer can create and track tickets
- Agent receives smart priority queue
- Category classification works
- Severity/priority prediction works
- SLA countdown works
- M2 knowledge retrieval works
- Grounded response includes sources
- M3 multi-agent workflow works
- Validation/confidence gate works
- Escalation works
- Jira create/update/sync works
- Email automation works
- Activity logs capture agent/integration events
- Ticket lifecycle is correct
- Analytics reflects ticket events
- Admin controls work
- RBAC is enforced
- Customer isolation is verified
- AI failures degrade safely
- End-to-end ticket resolution passes

29. Final Architecture



Final recommendation: for the full SupportPilot enterprise blueprint, use seven logical workspaces inside one frontend. For the simpler internship scope, start with the three mandatory roles — Customer, Agent and Admin — and add Lead, Knowledge Manager, Engineer/Agent Operator and Manager through RBAC as the project expands. This matches both the simplified project brief and the detailed technical blueprint. ■filecite■turn4file10■ ■filecite■turn4file3■